DCN CEP REPORT
CAMPUS NETWORKING PROJECT
DATA COMMUNICATION AND NETWORKS (EL-3002)
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Campus Network Design, Configuration, and Implementation

1. Introduction

This lab project focuses on designing and implementing a **multi-building campus network** using fundamental networking concepts such as **VLAN segmentation**, **DHCP services**, **LAN/WAN connectivity**, and **routing on multilayer switches and routers**. The objective is to design a scalable, secure, and efficient network for multiple departments across different buildings, interconnected through a main campus core and linked to a remote branch campus via a WAN connection.

The project simulates a real-world enterprise network of FAST university where multiple departments require resource sharing, controlled communication, and centralized server access.

2. Objectives

- To design a structured network topology for a university-style campus with multiple buildings.
- To implement **VLANs** for departmental isolation and traffic management.
- To configure **DHCP services** for automated IP addressing within each VLAN.
- To establish **inter-VLAN routing** using a Layer 3 switch.
- To set up **LAN connections** inside buildings and a **WAN connection** between campuses.
- To deploy network services such as **FTP, Web, and Email servers**.
- To ensure full end-to-end connectivity between main and branch campuses.

3. Network Overview

Main Campus Components:

- Core Router (2911)
- Main L3 Switch (3560) for Inter-VLAN Routing
- Building Access Switches (2960)
- Multiple VLANs for each department
- Centralized Servers:
- PCs and Printers per department

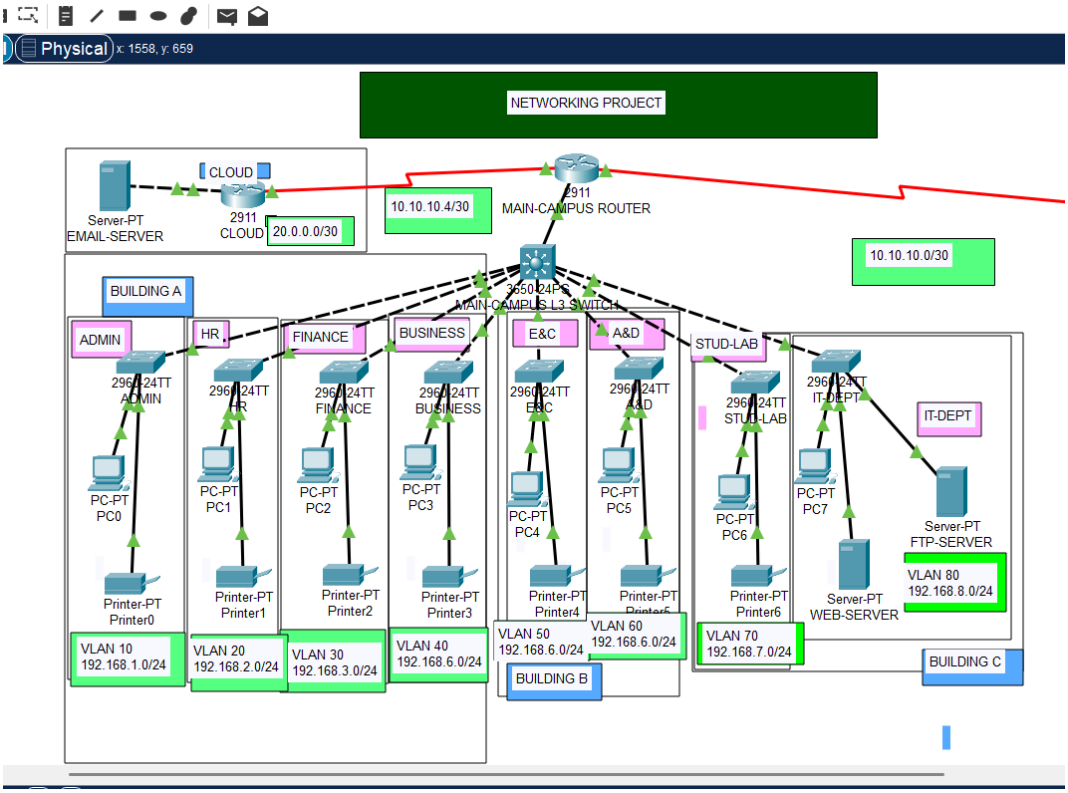
Branch Campus Components:

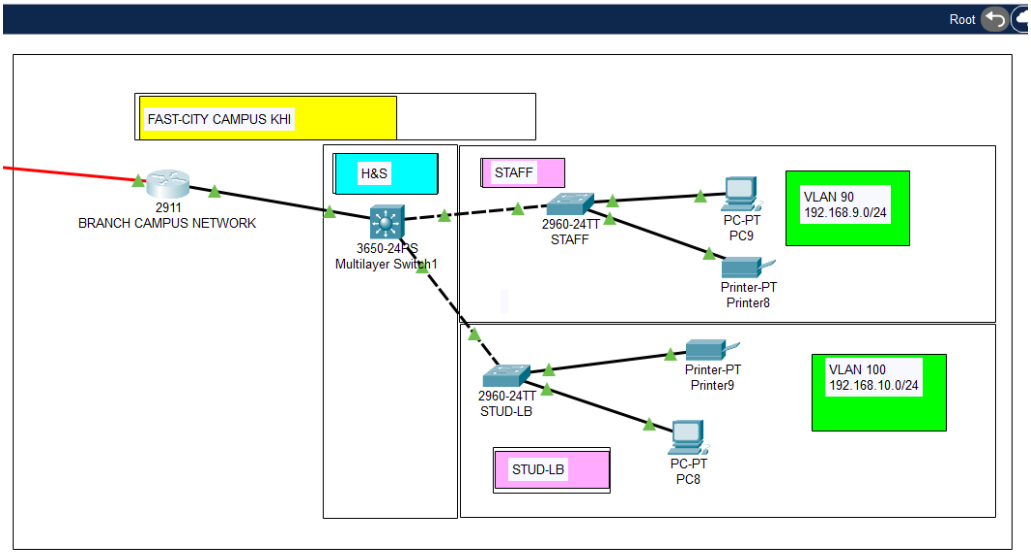
- Branch Router (2911) connected via WAN
- Multilayer Switch (3560)
- Departmental access switches (2960)
- End devices (PCs and printers)

WAN Connection:

- Main Campus Router (10.10.10.4/30)
- Branch Campus Router (10.10.10.0/30)

This WAN ensures full inter-campus communication.





4. VLAN Design

Department	VLAN	Subnet
Admin	10	192.168.1.0/24
HR	20	192.168.2.0/24
Finance	30	192.168.3.0/24
Business	40	192.168.4.0/24
E&C	50	192.168.5.0/24
A&D	60	192.168.6.0/24
Student Lab	70	192.168.7.0/24
IT Department	80	192.168.8.0/24

VLANs ensure security (traffic isolation), better performance (reduced broadcast domains) and administrative control. The L3 switch performs **Inter-VLAN Routing** using switched virtual interfaces (SVIs).

5. IP Addressing and DHCP Implementation

A centralized DHCP server dynamically assigns IPs to each VLAN.

DHCP Pools Created:

- DHCP scopes for all VLANs (10 to 80)
- Default gateway set as the corresponding SVI interface on the L3 switch
- DNS pointing to internal Email/Web servers

6. Inter-VLAN Routing

Inter-VLAN routing is implemented on the **3560 Multilayer Switch** using:

- SVI interfaces
- Routing enabled (`ip routing`)
- Trunk links between L3 switch and departmental access switches

This ensures devices in different VLANs can communicate through the L3 switch, while still maintaining traffic segmentation.

```
VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Gig1/0/1, Gig1/0/10,
Gig1/0/11, Gig1/0/12
                                               Gig1/0/13, Gig1/0/14,
Gig1/0/15, Gig1/0/16
                                               Gig1/0/17, Gig1/0/18,
Gig1/0/19, Gig1/0/20
                                               Gig1/0/21, Gig1/0/22,
Gig1/0/23, Gig1/0/24
                                               Gig1/1/1, Gig1/1/2,
Gig1/1/3, Gig1/1/4
10   VLAN0010                         active    Gig1/0/2
20   VLAN0020                         active    Gig1/0/3
30   VLAN0030                         active    Gig1/0/4
40   VLAN0040                         active    Gig1/0/5
50   VLAN0050                         active    Gig1/0/6
60   VLAN0060                         active    Gig1/0/7
70   VLAN0070                         active    Gig1/0/8
80   VLAN0080                         active    Gig1/0/9
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
```

7. LAN Design (Main Campus)

Each building has:

- One **2960 Access Switch**

- PCs and printers connected via access ports
- A trunk uplink toward the L3 switch
- Proper VLAN assignments
- Department-based segmentation

Printers and PCs receive IPs automatically through DHCP.

8. WAN Configuration (Main ↔ Branch Campus)

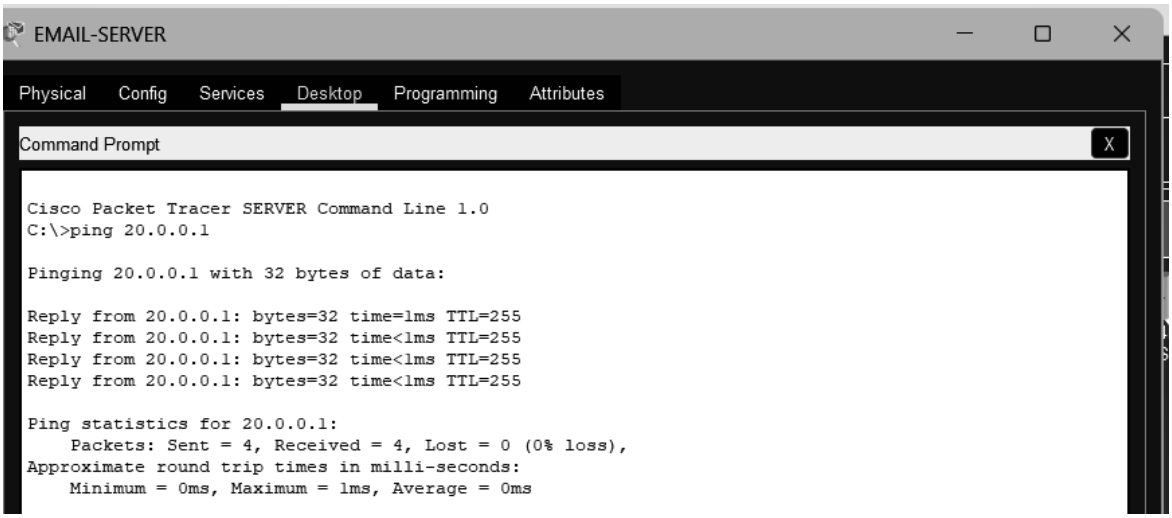
A serial/cloud link connects the main and branch campus routers.

Key configurations include:

- IP addressing of WAN interfaces (10.10.10.0/30)
- Static or dynamic routing (EIGRP/RIP/OSPF depending on design choice)
- Route propagation to ensure reachability between all VLANs across both campuses

This WAN connection enables:

- Remote access to centralized services
- Email/Web/FTP access from branch campus
- Administrative communication between campuses



9. Server Configurations

The main campus hosts multiple servers:

Email Server

- Provides internal SMTP/POP services
- Uses Cloud IP (20.0.0.0/30) for external connectivity

FTP Server

- Stores shared files for IT and Students

Web Server

- Hosts institutional web pages and internal applications

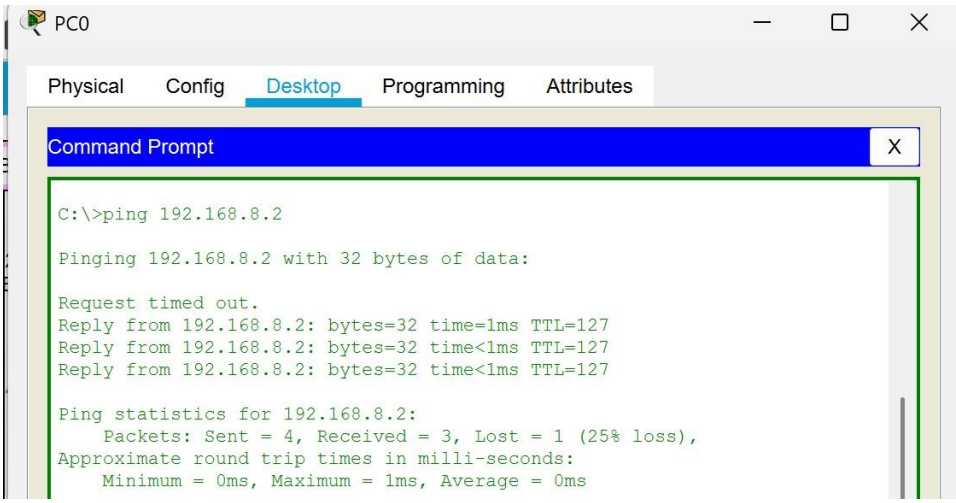
These servers are placed in the **IT VLAN (VLAN 80)** for centralized management.

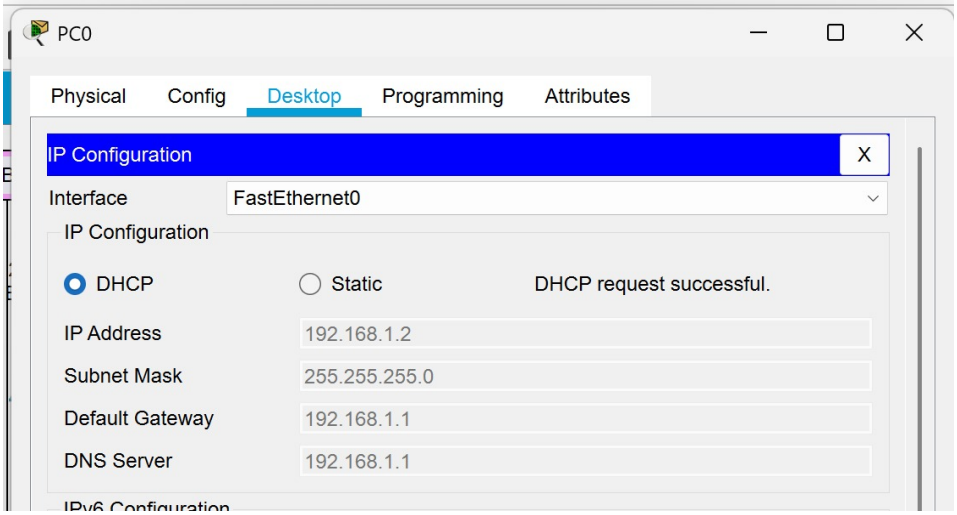
10. Testing & Verification

The following tasks were performed for validation:

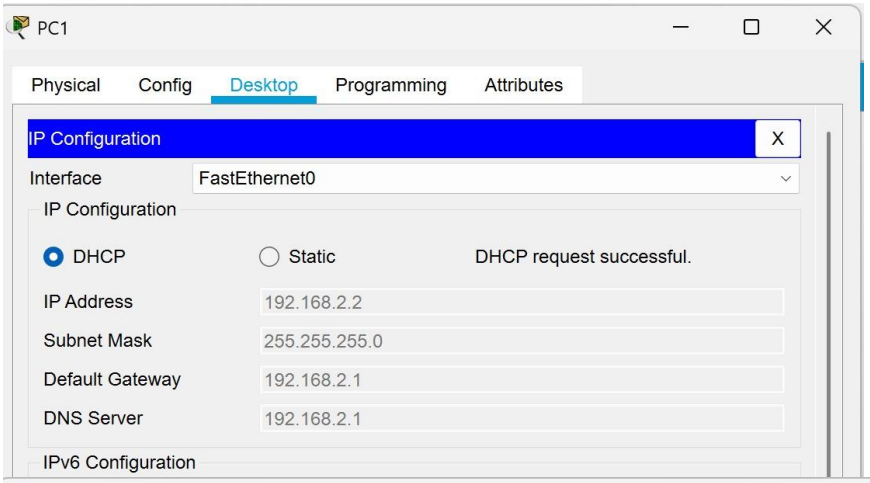
1. **Ping tests** across VLANs → Successful
2. **Ping from branch campus to main campus server** → Successful
3. **DHCP testing** on each PC → Successfully received IPs
4. **Printer accessibility** → Working within respective VLANs
5. **Web, FTP, and Email server tests** → Fully functional

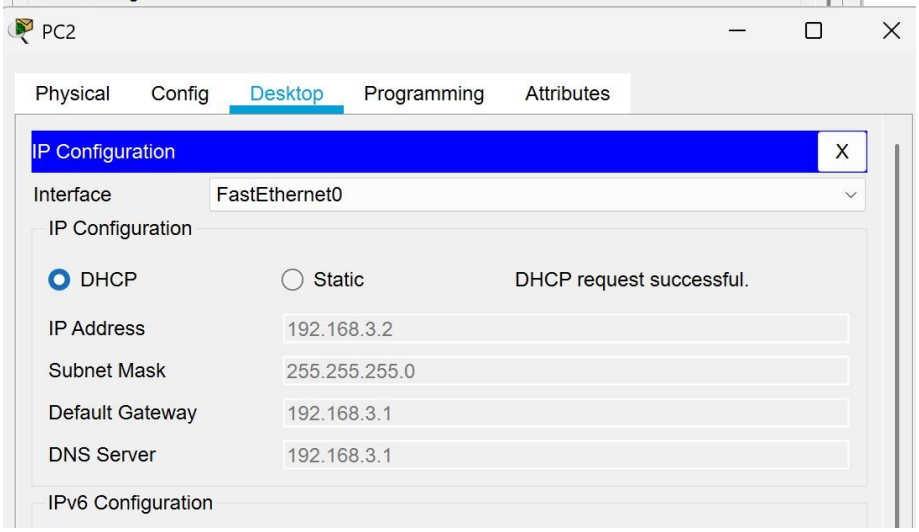
Ping for PC0:





Configurations for PC1 to PC9:





PC3

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static | DHCP request successful.

IP Address	192.168.4.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	192.168.4.1

IPv6 Configuration

DHCP | Auto Config | Static

PC4

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static | DHCP request successful.

IP Address	192.168.5.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.5.1
DNS Server	192.168.5.1

IPv6 Configuration

DHCP | Auto Config | Static

PC5

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static | DHCP request successful.

IP Address	192.168.6.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.6.1
DNS Server	192.168.6.1

IPv6 Configuration

PC6

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static | DHCP request successful.

IP Address	192.168.7.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.7.1
DNS Server	192.168.7.1

IPv6 Configuration

DHCP | Auto Config | Static

PC7

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static | DHCP request successful.

IP Address	192.168.8.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.8.1
DNS Server	192.168.8.1

IPv6 Configuration

PC8

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

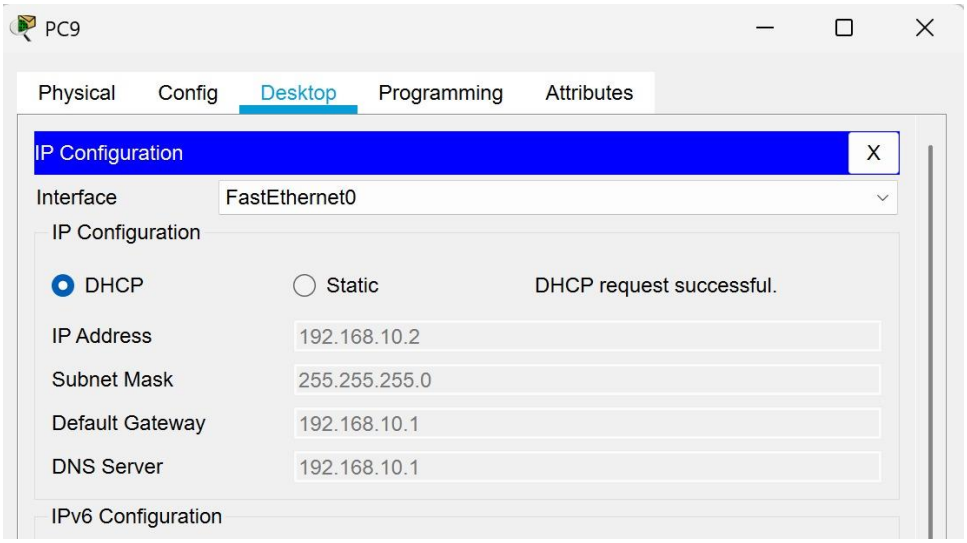
IP Configuration

DHCP | Static | DHCP request successful.

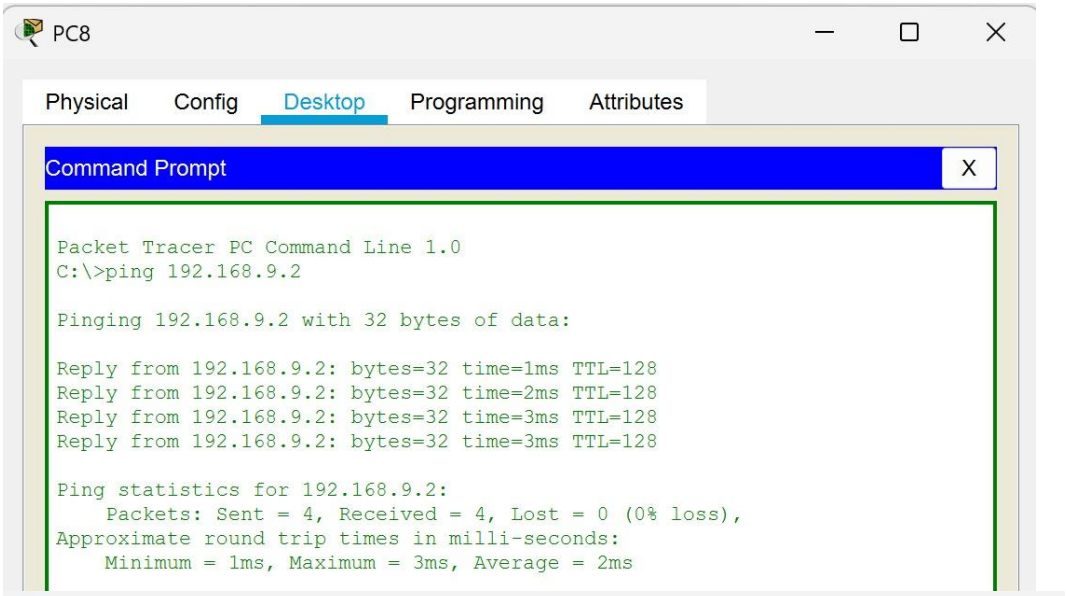
IP Address	192.168.9.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.9.1
DNS Server	192.168.9.1

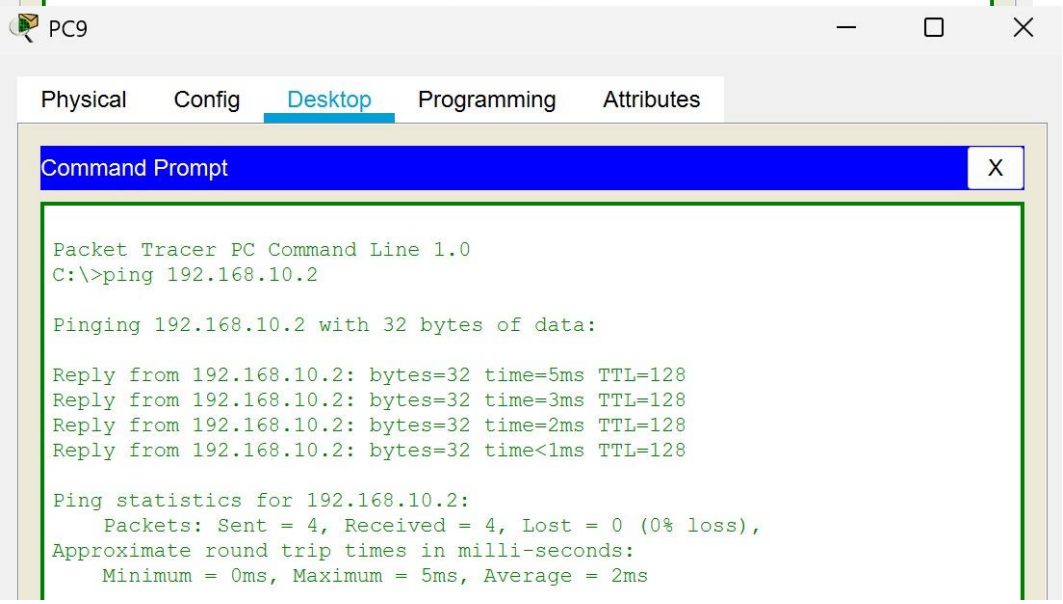
IPv6 Configuration

DHCP | Auto Config | Static



Ping for PC8 and PC9:





10. SIMULATION:

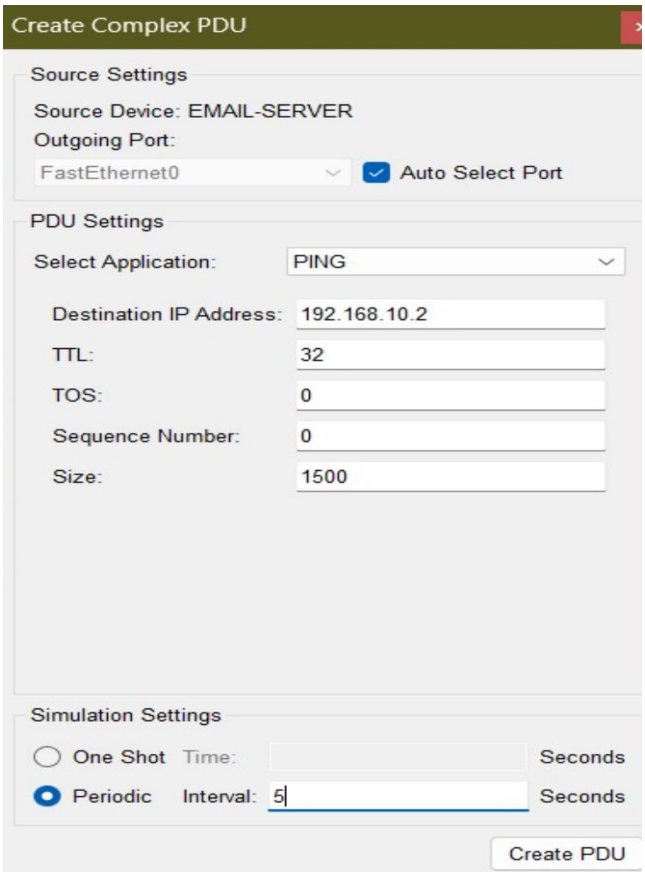


Figure 1: Message size is 1500 meeting the max of req of MTU of a router

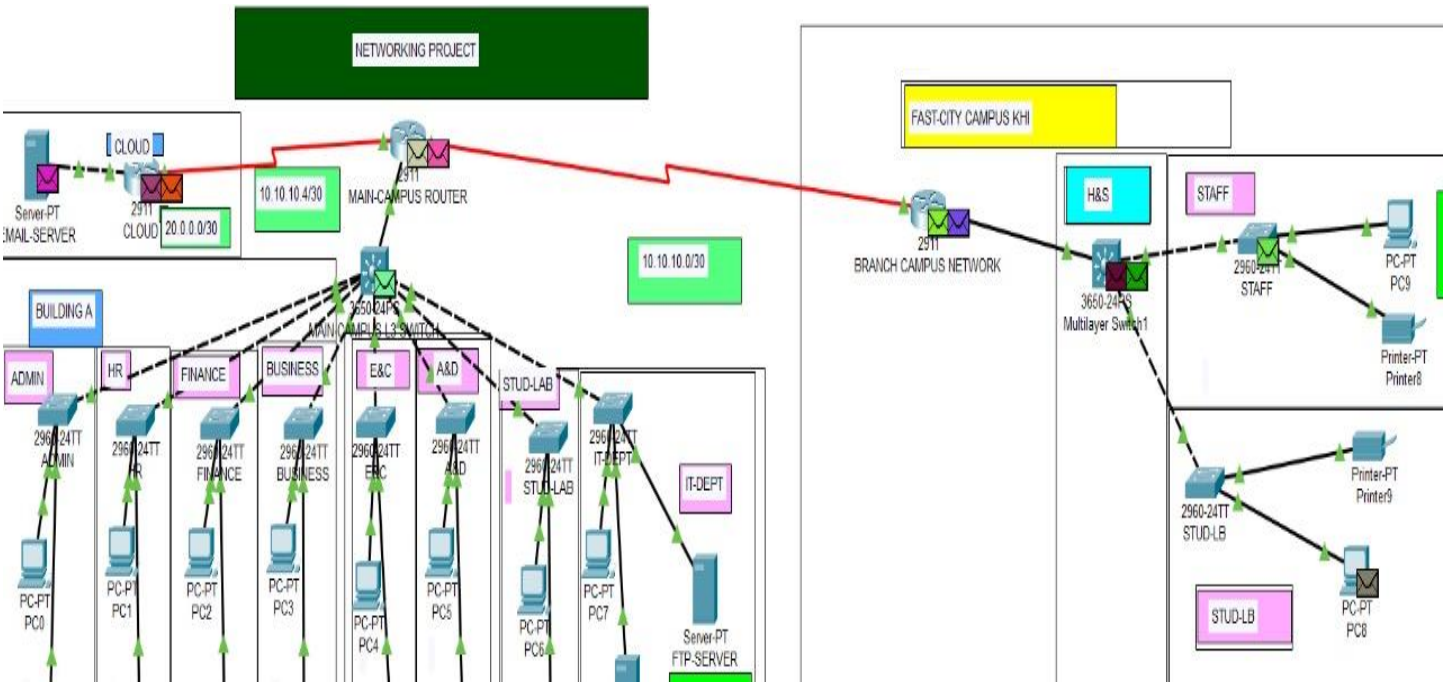


Figure 2: Packet was initiated at server and travelled to PC-8

11. Results

The designed network successfully met all requirements:

- All departments operate with isolated VLANs.
- DHCP assigns IPs without conflict.
- Inter-VLAN routing is operational.
- Servers are reachable from both campuses.
- WAN link allows full two-way campus communication.
- The network is scalable, secure, and efficient.

12. Conclusion

This lab project demonstrated the practical implementation of a multi-building campus network using professional networking techniques. By integrating VLAN segmentation, DHCP automation, inter-VLAN routing, LAN/WAN deployment, and centralized servers, the project effectively replicates a real-world enterprise environment.

The resulting design supports scalability, resource sharing, improved performance, and simplified network management. This lab reinforced theoretical concepts from the Data Communication & Networking course and provided hands-on experience with complex network architecture.